
Evaluating Offline-to-Online Selection Transfer under Fixed Decision Interfaces

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Abstract

Offline decision-making pipelines often select models using validation scores computed on collected data, then deploy the selected model through a fixed decision interface such as threshold alerts, hysteresis rules, budgeted top- k actions, residual-warning screens, or replenishment rules. We ask a narrow evaluation question: does the offline-selected model remain the deployed-utility winner after the same interface and deployment friction are applied? We propose an offline-to-deployed selection-transfer report card that records the offline-selected model, deployed-utility winner, agreement rate, deployed-suboptimal share, paired utility shortfall, and tie diagnostics. Across a synthetic zero-friction anchor, event-warning, traffic-alert, and EPA PM2.5 residual-warning tasks, with inventory as operational corroboration, models selected by offline validation scores become recurrently deployed-suboptimal in moderate-to-high friction regimes in the studied settings. This is not an online adaptation algorithm, a general off-policy evaluation method, or a universal offline decision-making benchmark suite; “online” refers to deployed-outcome evaluation under a specified fixed interface, not online model updating.

1. Introduction

Offline decision-making pipelines often select a model using validation scores computed on collected data, then deploy the selected model through a fixed decision interface that maps predictions or scores into actions. Examples include triggering an alert when $p_t > \tau$, changing state only if the score moves by at least δ , issuing the top- k alerts under a fixed budget, applying a residual-warning screen, or applying a replenishment rule. When the deployed interface in-

cludes switching or adjustment friction, the model selected by the offline score need not remain the deployed-utility winner. We do not tune the interface per model: the same specified decision interface is applied to every candidate model before deployed-utility scoring.

This paper studies offline-to-deployed selection transfer as a reporting diagnostic. By selection transfer we mean whether the model selected by an offline validation score remains the best model under deployed utility after a fixed decision interface and friction model are applied. This is not an argument that forecast-quality metrics are invalid for prediction, nor a proposal for a single deployed metric. Deployed utility depends on a specified interface, simulator, and friction model. Our claim is narrower: when offline validation is used as deployment-facing model-selection advice, the evaluation should report whether that selection transfers to deployed utility under the specified fixed interface. Throughout, “online” means deployed-outcome evaluation under the specified interface, not online adaptation or model updating.

This paper makes three focused contributions.

- Selection-transfer diagnostic. We formulate offline-to-deployed selection-transfer robustness: whether the model selected by an offline validation score remains the best model under deployed utility after a fixed decision interface and friction model are applied.
- Report-card artifact. We provide a lightweight report card recording offline-selected models, deployed-utility winners, agreement rates, deployed-suboptimal shares, paired utility shortfalls, and tie/uncertainty diagnostics.
- Empirical audit. Across prediction-to-decision tasks, we find recurrent selection failures in moderate-to-high friction regimes, while retaining zero-friction anchors, failed-gate variants, and tie audits to bound the interpretation.

Setup. For each candidate model $m \in \mathcal{M}$, an offline validation score $S_{\text{off}}(m)$ selects $m_{\text{off}} = \arg \min_m S_{\text{off}}(m)$. We use the loss convention for S_{off} , with lower values better. A fixed decision interface I_θ maps model outputs to actions, $a_{1:T}^m = I_\theta(\hat{y}_{1:T}^m)$, and the same specified interface is applied to every candidate model. We evaluate deployed utility $U_\kappa(m; I_\theta)$ under friction κ , with larger values better. The deployed-utility winner is $m_{\text{dep}} = \arg \max_m U_\kappa(m; I_\theta)$. Selection transfer is preserved when $m_{\text{off}} = m_{\text{dep}}$; other-

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Offline-selected models can become deployed-suboptimal.

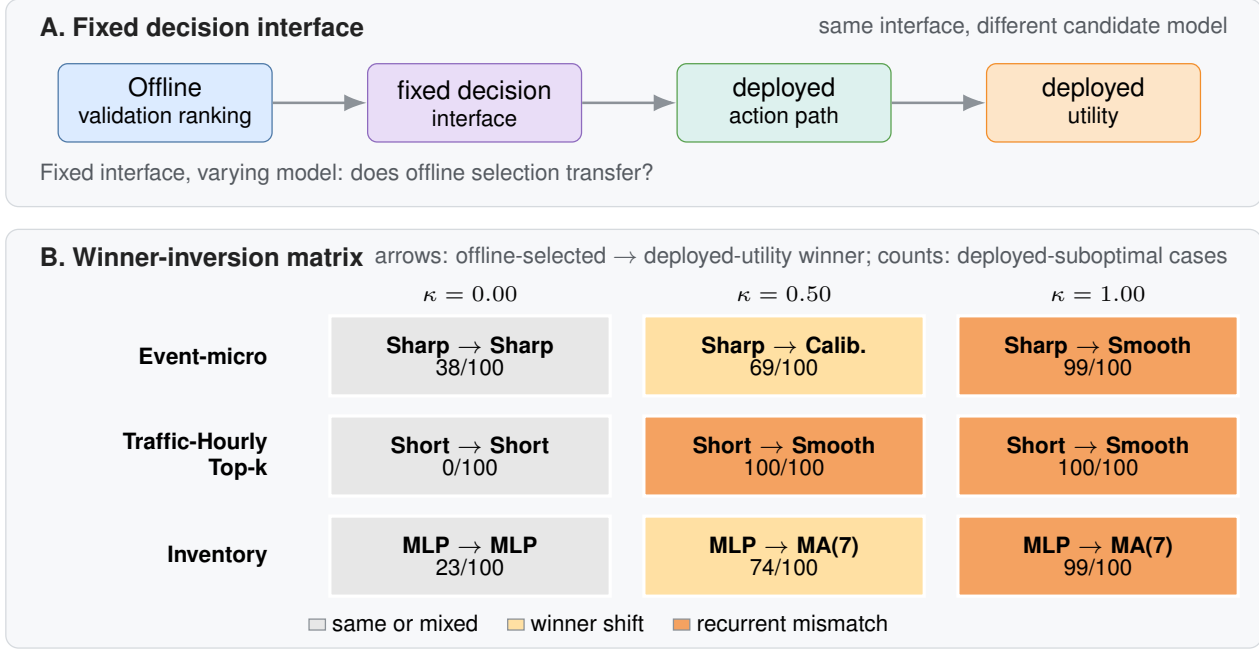


Figure 1. Offline model selection need not transfer through a fixed decision interface. Panel A shows the selection-transfer setting: models are ranked by an offline validation score, passed through the same fixed decision interface, and evaluated by deployed utility under friction. Panel B reports aggregate offline-selected and deployed-utility winners across representative task families and friction levels. Arrows show offline-selected model $\rightarrow$ deployed-utility winner; counts denote evaluation units in which the offline-selected model is deployed-suboptimal. Gray zero-friction cells are included for completeness rather than interpreted as friction-causal anchors; the clean zero-friction anchors are the synthetic sanity row in Table 1 and the Traffic-Hourly Top- k zero-friction cell.

wise the offline-selected model incurs deployed shortfall $\Delta_\kappa = U_\kappa(m_{\text{dep}}; I_\theta) - U_\kappa(m_{\text{off}}; I_\theta)$. All deployed outcomes are reported as utilities or utility shortfalls, so larger U_κ is better and $\Delta_\kappa > 0$ means the offline-selected model is deployed-suboptimal. Near-ties are handled by the pre-specified tie-breaking rule and by the ϵ -tie audit reported in the appendix.

We use Q1/Q2 only to separate mechanism from model-selection evidence. Q1 fixes the proposed action path and varies the interface as an exact-control check. In Q2, the interface is fixed and the candidate prediction model varies; this is the main model-selection question.

2. Results

2.1. Selection transfer under fixed decision interfaces

This corresponds to the paper’s Q2 question: the interface is fixed and the candidate model varies. The report card ranks candidate models by an offline validation score, applies the same fixed decision interface to every candidate, evaluates deployed utility across a friction grid, and reports whether the offline-selected model remains the deployed-utility winner using transfer rate, deployed-suboptimal share, paired

shortfall, and tie diagnostics. Table 1 is the output format of this diagnostic. The zero-friction synthetic row is a sanity anchor. Event-micro and Traffic-Hourly provide the core prediction-to-decision Q2 evidence under two different fixed-interface structures, the EPA residual-warning audit adds one additional prediction-to-decision row, and inventory is included only as operational corroboration. Figure 1 visualizes the main winner-inversion evidence across the prediction-to-decision and corroboration rows.

Event-micro is the primary selection-transfer failure. Brier and log-loss act as offline validation scores, while threshold and hysteresis rules are fixed decision interfaces. In the report card, the offline-selected model remains Reactive sharp, but the deployed-utility winner changes to Calibrated baseline at friction 0.50 and Lagged smoother at friction 1.00. The deployed-utility consequence is recurrent: the offline-selected model is deployed-suboptimal in 69/100 and 99/100 seeds at these two friction levels. The result is not confined to the canonical threshold or Brier ranking: at frictions 0.50/1.00, deployed-suboptimality remains 77/100 and 98/100 under an alternate threshold, 74/100 and 99/100 under log-loss reranking, and 55/100 and 82/100 under a hysteresis interface, where the direction is preserved with

Table 1. Offline-to-deployed selection-transfer report card. Each row fixes a task family, decision interface, and friction level, then asks whether the model selected by the offline validation score remains the deployed-utility winner after the same interface is applied. “Offline sel.” denotes the offline-selected model; “Deploy winner” denotes the best model under deployed utility; “Transfer” is the fraction of evaluation units where the offline-selected model also wins under deployed utility. “Subopt.” reports deployed-suboptimal share/count; intervals are paired-bootstrap intervals for the mean deployed shortfall.

Domain	Interface	κ	Offline sel.	Deploy winner	Transfer	Subopt.	Mean shortfall	95% CI	Role
Synthetic	zero-friction anchor	0.00	Naive last	Naive last	1.00	0.00 (0/20)	0.000	[0.000, 0.000]	Sanity
Event-micro	threshold $\tau=0.55$	0.50	Reactive sharp	Calibrated	0.31	0.69 (69/100)	0.011	[0.009, 0.014]	Primary
Event-micro	threshold $\tau=0.55$	1.00	Reactive sharp	Smoother	0.01	0.99 (99/100)	0.057	[0.052, 0.063]	Primary
Traffic Top- k	budget $k=249$	0.50	Reactive short	Smoother	0.00	1.00 (100/100)	7.016	[6.756, 7.283]	Second
Traffic Top- k	budget $k=249$	1.00	Reactive short	Smoother	0.00	1.00 (100/100)	24.651	[24.226, 25.079]	Second
EPA AQS	PM2.5 residual warn- ing	1.00	Reactive lag-1	Long smoother	0.00	0.87 (720/825)	17.939	[16.560, 19.342]	Third
Inventory	replenishment	1.00	Small MLP	MA(7)	0.01	0.99 (99/100)	1.029	[0.946, 1.111]	Corrob.

smaller magnitude.

Traffic-Hourly Top- k is a prediction-to-decision task: forecasts are ranked by an offline validation score, but the fixed interface deploys only the top- k actions under a budget. In the selected Top- k setting ($k = 249$), Reactive short is both the offline-selected model and the deployed-utility winner at zero friction, with exact transfer 1.0. Once switching costs are introduced, however, the offline-selected model remains Reactive short while the deployed-utility winner shifts to Lagged smoother, and deployed-suboptimality reaches 100/100 seeds at both friction 0.5 and friction 1.0. Additional Traffic checks support this reading most strongly for the Top- k interface family: across $k \in \{125, 249, 500\}$, zero-friction rows remain aligned and positive-friction rows keep the reactive offline-selected model deployed-suboptimal; predefined time-of-day and relative-rank variants are kept in the appendix rather than used to broaden the main claim.

EPA PM2.5 adds a broader residual-warning audit row without changing the evidence hierarchy. In expanded EPA AQS PM2.5 residual warning, Reactive lag-1 remains the offline-selected model, while Long smoother becomes the deployed-utility winner at $\kappa = 1.00$. We treat its units as report-card evaluation cells rather than independent iid samples; the appendix reports the full residual-candidate audit, including appendix-only Green support and failed-gate context.

An ϵ -tie audit gives the same qualitative reading for the main prediction-to-decision rows: Traffic-Hourly Top- k and Event-micro at $\kappa \in \{0.50, 1.00\}$, and expanded EPA at $\kappa = 1.00$, remain deployed-suboptimal under relative tie thresholds up to $\epsilon_{\text{rel}} = 0.005$; inventory remains corroborative and lower-margin secondary checks are reported only in the appendix.

Inventory provides operational prediction-to-decision corroboration, but we keep it bounded because the interface and utility are domain-specific. At friction 0.50 and 1.00, the offline-selected Small MLP is deployed-suboptimal in 74/100 and 99/100 seeds, respectively, under the fixed replenishment interface; the report card includes only the high-friction inventory row to keep this role bounded.

Q1 mechanism support. Q1 explains why divergence can arise; Q2 is the actual benchmark-selection claim. Holding the proposed action path fixed, synthetic and inventory show that frictional interfaces can separate proposed and realized actions. Synthetic yields an increasing proposal-versus-realization gap away from zero friction, while inventory shows a threshold pattern in which sufficiently high friction favors tempered execution. Q2 then shows when that mechanism produces deployed misselection under one fixed interface.

Appendix robustness. Appendix checks surface robustness without changing the evidence hierarchy: event-micro includes alternate-threshold, log-loss, and hysteresis-interface checks; Traffic-Hourly includes Top- k interface-family, predefined time-of-day slice, and relative-rank checks; residual-candidate audits retain Green support and failed-gate context; and expanded inventory and load-following remain fairness or appendix-only context. These checks are consistent with the main qualitative conclusion: under a fixed interface, an offline-selected model can remain validation-best while becoming recurrently deployed-suboptimal in moderate-to-high friction regimes.

Why offline ordering need not be preserved. Offline validation scores evaluate their specified validation objective, not the composed object obtained after model outputs are mapped to actions and executed through a frictional interface. If a candidate model’s offline validation advantage is smaller than the extra friction-induced loss created by its implied action path, the deployed ordering can reverse even under a fixed interface. This is why Q1 is useful as mechanism support and why Q2 is the actual selection-transfer question.

3. Discussion, Related Work, and Limitations

Offline decision-making evaluation protocols. Our work is closest to the evaluation-protocol side of offline decision-making: before asking how to adapt online, we ask whether the model selected offline remains preferred after a specified deployment interface is applied. Policy comparison and off-policy evaluation are adjacent, but our object is narrower than estimating the value of arbitrary policies: we audit whether offline validation selection transfers through a fixed decision interface and friction model. Unlike general OPE, our fixed-interface replay setting does not estimate arbitrary counterfactual policies from logged propensities; it evaluates a fixed candidate set through the same specified decision interface on held-out outcomes.

Forecasting benchmarks and proper scoring. Forecasting benchmarks and evaluation frameworks now span live AI forecasting, broad time-series archives, foundation-model evaluations, text-conditioned forecasting, and standardized TSFM evaluation pipelines (Karger et al., 2025; Yang et al., 2025; Aksu et al., 2024; Godahewa et al., 2021; Das et al., 2024; Williams et al., 2025; Li et al., 2025; Goktas et al., 2026; Zhao et al., 2025). These works compare forecasting capability, datasets, model families, or evaluation pipelines, and proper scoring rules remain the right object when the target is predictive quality (Gneiting and Raftery, 2007; Murphy, 1993). Our question is complementary: when a prediction benchmark is interpreted as deployment-facing model-selection advice under a fixed decision interface, should it also report whether offline validation selection transfers to deployed utility?

Decision-focused learning and benchmark uncertainty. This is distinct from decision-maker-perspective forecast evaluation, decision-focused learning, and predict-then-optimize, which either define forecast value or modify learning or optimization objectives for downstream decisions (Raeth and Ludwig, 2025; Donti et al., 2017; Elmachtoub and Grigas, 2022; Mandi et al., 2024). Here, candidate models and the interface are fixed; the contribution is a reporting diagnostic for whether offline model selection survives the composed model-output-to-interface-to-utility

evaluation. Our uncertainty reporting follows the spirit of benchmark-variance and leaderboard-interpretation work (Bouthillier et al., 2021; Longjohn et al., 2025; Shchur et al., 2025; Brigato et al., 2026), but reports selection transfer and deployed shortfalls rather than aggregate predictive leaderboard uncertainty. Deployment frictions such as switching costs, adjustment costs, and operational churn can change which model is preferable after actions are executed; our report card makes this effect visible at the model-selection level.

Discussion and limitations. Our claim is evaluative and deliberately narrow. We do not claim that forecasting metrics are invalid for prediction, and we do not seek a single reporting rule for all applications. We use low, moderate, and high only as shorthand for the lower, middle, and upper parts of each domain’s tested friction grid; these labels are not intended as a universal calibration across domains. Event-micro and Traffic-Hourly provide the core prediction-to-decision evidence, the EPA residual-warning audit adds one additional report-card row, and inventory provides operational corroboration, but this is still not a comprehensive benchmark collection. Broader inventory sweeps are reported in the appendix, and load-following is secondary; the strongest main-text evidence comes from the moderate-to-high friction settings studied here. Our contribution is not a new policy-learning algorithm, a general OPE method, an online adaptation algorithm, a new forecasting model, a universal deployed metric, or a universal offline decision-making benchmark suite. It is a narrow reporting recommendation: when offline validation is interpreted as model-selection advice for a fixed deployment interface, it should also report whether offline selection transfers to deployed utility. Under this interpretation, inventory and load-following remain corroborative or secondary context rather than standalone statistical evidence for a broader population-level claim.

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Minimum reporting standard for offline-to-deployed selection transfer. For each task family, the report-card artifact records:

1. task definition, seed count, and evaluation split;
2. offline validation score and tie-breaking rule;
3. fixed decision interface and interface parameters;
4. friction grid, including a zero-friction anchor and its status when applicable;
5. candidate model families and representative selection rule;
6. split/seed manifest;
7. offline-selected model and deployed-utility winner for each domain $\times$ interface $\times$ friction cell;
8. agreement rate and deployed-suboptimal share/count of the offline-selected model;
9. mean and median paired deployed-utility shortfalls with uncertainty intervals;
10. uncertainty and ϵ -tie audit;
11. mixed or failed-gate variants, when applicable.

In appendix tables inherited from generated artifacts, “Forecast-side winner” or “Forecast winner” should be read as the offline-selected model, and “Deployed winner” as the deployed-utility winner.

Mixed and failed-gate variants. We retain mixed and failed-gate variants as audit evidence rather than promoting them into the main report card. This reduces the risk that the report card is read as a collection of selected successes: variants that fail the zero-friction alignment check, remain lower-margin under tie diagnostics, or show only partial recurrence are kept as appendix context or excluded from main-text claims.

Table A1. Mixed and failed-gate variants retained as audit context. These variants are not promoted into the main report card because they either fail the clean zero-friction alignment check, show lower-margin or tie-sensitive behavior, or provide only partial recurrence.

Variant	Gate outcome	Interpretation	Placement
Traffic daytime $\kappa = 0.5$	Partial	Lower-margin, directionally consistent	Appendix only
Inventory zero/low friction	Mixed	Not a main zero-friction anchor	Appendix only
Expanded inventory	Mixed	Fairness context only	Appendix only
Load-following	Tie-sensitive	Directional secondary check	Appendix only
Load-alert screening	Yellow/Red	Daily peak-hour Yellow; residual/four-week Red	Screening artifacts
Residual-candidate audit	Mixed	Expanded EPA/Capital/Wikimedia pass; NYC/M5 fail zero-friction alignment	Appendix C

These variants are useful because they separate friction-induced selection-transfer failure from other sources of mismatch, such as objective mismatch, weak margins, or insufficient seed support. We therefore use them to bound the interpretation rather than to strengthen the main claim.

A. Event-Warning Decision Task Robustness

Because event-micro provides the paper’s main prediction-to-decision selection-transfer evidence, this appendix records the first domain-specific robustness package for that benchmark. We construct a minimal binary-event setting evaluated over 100 seeds, in which each candidate prediction model emits event probabilities, a fixed thresholding rule maps probabilities to actions, and switching friction penalizes action changes. The main text uses Brier as the canonical offline validation score.

This appendix records the canonical six-row Brier summary plus an alternate-threshold check, a log-loss reranking check, and a second-interface hysteresis check.

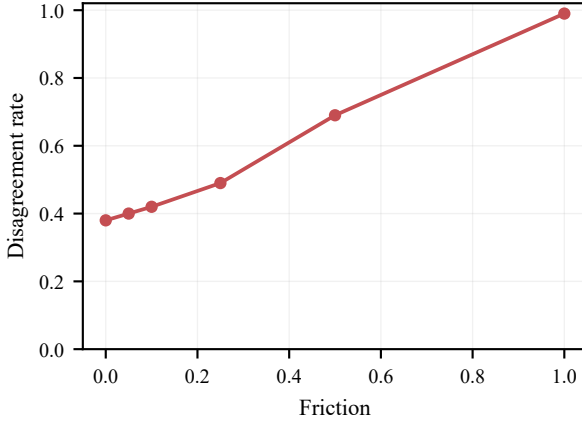


Figure A1. Event-micro canonical summary. Left: disagreement between offline and deployed-utility model selection increases with friction. Right: canonical fixed-threshold selection summary; shortfall is the mean paired deployed-utility shortfall of the offline-selected model.

Threshold	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
tau=0.55	0.00	Reactive sharp	Reactive sharp	0.62	0.003	0.000	38/100
tau=0.55	0.50	Reactive sharp	Calibrated baseline	0.31	0.011	0.008	69/100
tau=0.55	1.00	Reactive sharp	Lagged smoother	0.01	0.057	0.053	99/100
tau=0.50	0.00	Reactive sharp	Reactive sharp	0.56	0.002	0.000	44/100
tau=0.50	0.50	Reactive sharp	Calibrated baseline	0.23	0.014	0.011	77/100
tau=0.50	1.00	Reactive sharp	Lagged smoother	0.02	0.061	0.059	98/100

Table A2. Alternate-threshold robustness for the event-micro benchmark. The same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists when the fixed threshold is changed from $\tau = 0.55$ to $\tau = 0.50$.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive sharp	Reactive sharp	0.62	0.002	0.000	38/100
0.50	Reactive sharp	Calibrated baseline	0.26	0.013	0.010	74/100
1.00	Reactive sharp	Lagged smoother	0.01	0.060	0.057	99/100

Table A3. Log-loss robustness for the event-micro benchmark at the canonical threshold. The same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists when offline validation ranking is recomputed with log loss instead of Brier.

Interface	Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
Hysteresis threshold	0.00	Reactive sharp	Reactive sharp	0.55	0.003	0.000	45/100
Hysteresis threshold	0.50	Reactive sharp	Calibrated baseline	0.45	0.005	0.001	55/100
Hysteresis threshold	1.00	Reactive sharp	Lagged smoother	0.18	0.022	0.020	82/100

Table A4. Second-interface robustness for the event-micro benchmark. Under a fixed hysteresis-threshold interface with $\delta = 0.05$, the same qualitative selection-transfer mismatch and deployed-suboptimality pattern remains visible, though the magnitudes are smaller than under the canonical fixed threshold.

Across this appendix robustness package, the same qualitative selection-transfer mismatch and recurrent deployed-suboptimality pattern persists in a minimal prediction-to-decision event-micro benchmark evaluated over 100 seeds.

A.1. Interface-Aware Event-Micro Stats Addendum

This addendum records interface-specific recurrence and gap summaries for the hysteresis-threshold check without changing the main cross-domain recurrence package.

Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Fixed threshold	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Fixed threshold	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Hysteresis threshold	0.50	100	0.45	0.55 (55/100)	[0.452, 0.644]	Wilson
Hysteresis threshold	1.00	100	0.18	0.82 (82/100)	[0.733, 0.883]	Wilson

Table A5. Interface-aware recurrence and share uncertainty for event-micro deployed-suboptimality. The canonical fixed threshold shows stronger recurrence, while hysteresis preserves the same direction with smaller magnitudes.

The aggregate recurrence table reports the seed-level pattern directly, replacing the dense seed-strip visualization.

Interface	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Fixed threshold	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Fixed threshold	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Hysteresis threshold	0.50	0.005	[0.004, 0.006]	0.001	[0.000, 0.003]
Hysteresis threshold	1.00	0.022	[0.019, 0.026]	0.020	[0.015, 0.025]

Table A6. Interface-aware bootstrap intervals for the event-micro deployed gap. Positive intervals remain visible under both interfaces, with smaller gaps under hysteresis.

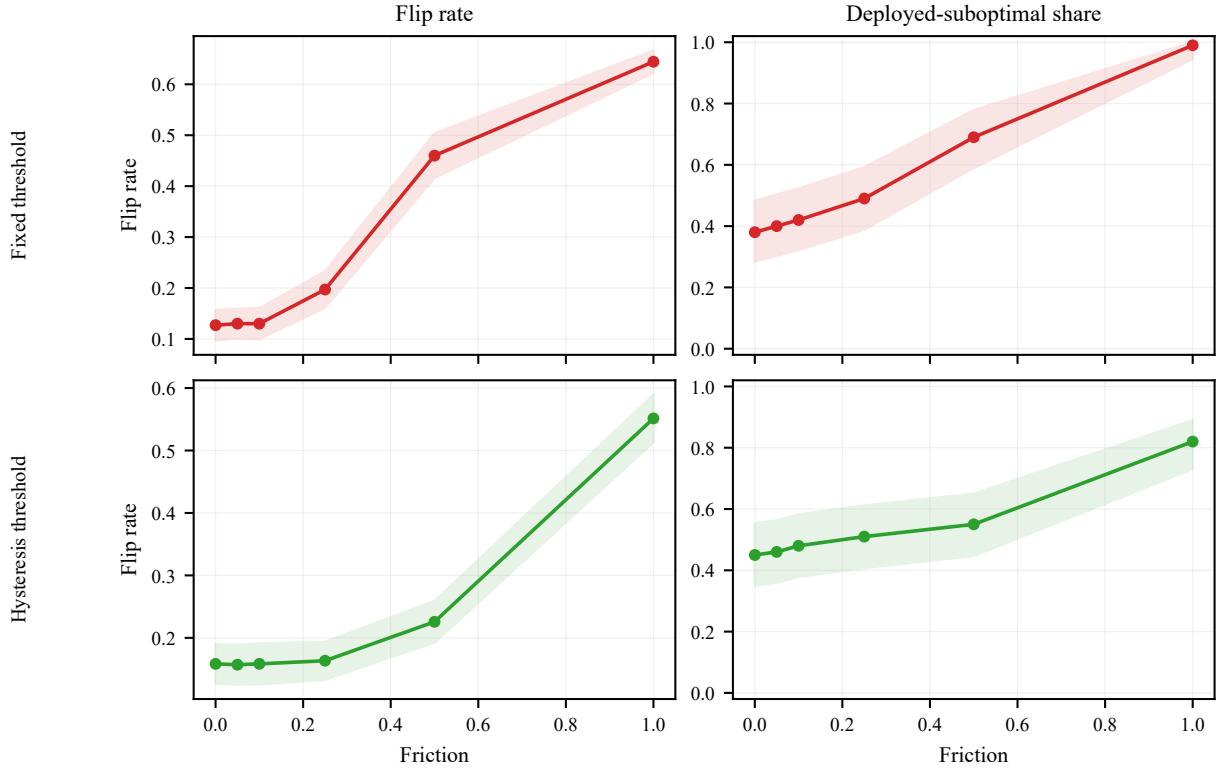


Figure A2. Event-micro robustness across fixed-threshold and hysteresis interfaces. Rows denote interfaces and columns denote disagreement and deployed-suboptimality metrics; hysteresis preserves the direction with attenuated magnitude.

B. Traffic Fixed-Budget Alert Robustness

B.1. Fixed-Budget Top-k Alert

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Lagged smoother	0.00	7.016	6.977	100/100
1.00	Reactive short	Lagged smoother	0.00	24.651	24.473	100/100

Table A7. Traffic-Hourly Top- k prediction-to-decision check under a fixed-budget interface. Agreement is strongest at zero friction and weakens at higher friction, where the deployed-utility winner shifts away from the offline-selected model.

In the selected Top- k setting, Reactive short is the offline-selected model and the deployed-utility winner at zero friction. At frictions 0.5 and 1.0, the deployed-utility winner shifts to Lagged smoother even though Reactive short remains Brier-best and continues to win under the offline validation score. The qualitative reading is the same as in the main text: under a fixed interface, the more reactive validation-optimal family need not remain deployment-optimal once switching is penalized.

B.2. Top- k Interface-Family Sweep

k	κ	Forecast winner	Deployed winner	Subopt.	Mean gap	95% CI	Median gap	Verdict
125	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
125	0.50	Reactive short	Calibrated baseline	100/100	3.794	[3.584, 4.012]	3.725	Pass
125	1.00	Reactive short	Calibrated baseline	100/100	15.986	[15.639, 16.339]	15.955	Pass
249	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
249	0.50	Reactive short	Lagged smoother	100/100	7.016	[6.756, 7.283]	6.977	Pass
249	1.00	Reactive short	Lagged smoother	100/100	24.651	[24.226, 25.079]	24.473	Pass
500	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
500	0.50	Reactive short	Lagged smoother	100/100	14.314	[14.048, 14.574]	14.383	Pass
500	1.00	Reactive short	Lagged smoother	100/100	37.985	[37.521, 38.438]	38.080	Pass

Table A8. Traffic-Hourly Top- k interface-family sweep. The canonical $k = 249$ rows are reproduced from the recovered time-level Traffic source, and non-canonical $k = 125$ and $k = 500$ rows show the same qualitative pattern: zero-friction alignment and recurrent positive-friction deployed-suboptimality of the reactive offline-selected model. Confidence intervals are paired-bootstrap intervals for the mean deployed shortfall.

Across $k \in \{125, 249, 500\}$, Reactive short remains both the offline-selected model and deployed-utility winner at zero friction. At positive friction, Reactive short remains validation-best but becomes deployed-suboptimal in all matched seeds, with positive paired deployed shortfalls and confidence intervals bounded away from zero.

B.3. Predefined Time-of-Day Slice Check

Slices are fixed before outcome computation as overnight, daytime, and evening blocks. Switching costs are computed on the full 168-hour timeline and utilities are then aggregated over the slice time points, so the interface state is not reset within each slice.

Slice	κ	Forecast winner	Deployed winner	Subopt.	Mean gap	95% CI	Median gap	Verdict
Overnight	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Overnight	0.50	Reactive short	Lagged smoother	100/100	13.981	[13.681, 14.279]	14.087	Pass
Overnight	1.00	Reactive short	Lagged smoother	100/100	29.858	[29.289, 30.430]	30.141	Pass
Daytime	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Daytime	0.50	Reactive short	Calibrated baseline	66/100	1.731	[1.385, 2.085]	1.205	Partial
Daytime	1.00	Reactive short	Calibrated baseline	100/100	16.049	[15.323, 16.762]	16.087	Pass
Evening	0.00	Reactive short	Reactive short	0/100	0.000	[0.000, 0.000]	0.000	Aligned
Evening	0.50	Reactive short	Calibrated baseline	100/100	10.086	[9.717, 10.444]	10.027	Pass
Evening	1.00	Reactive short	Calibrated baseline	100/100	33.269	[32.649, 33.873]	33.288	Pass

Table A9. Predefined Traffic-Hourly time-of-day slice check. Slices are fixed before outcome computation. Overnight and evening reproduce the positive-friction mismatch pattern at both positive-friction levels; daytime is directionally consistent but lower-margin at $\kappa = 0.5$. We therefore treat this slice check as appendix-only support. Confidence intervals are paired-bootstrap intervals for the mean deployed shortfall.

The slice check is supportive but weaker than the interface-family sweep. Overnight and evening pass at both positive-friction levels, while daytime passes at $\kappa = 1.0$ and remains directionally positive at $\kappa = 0.5$ but falls below the pre-specified deployed-suboptimality threshold. We therefore use this result only as appendix context rather than as a stronger main-text claim.

B.4. Relative-Rank Traffic Variant

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal seeds / total
0.00	Reactive short	Reactive short	1.00	0.000	0.000	0/100
0.50	Reactive short	Calibrated baseline	0.00	10.103	10.122	100/100
1.00	Reactive short	Calibrated baseline	0.00	25.205	25.231	100/100

Table A10. Traffic-Hourly relative-rank appendix variant. The same qualitative separation between offline and deployed-utility selection appears under fixed-budget set action.

As an additional Traffic appendix variant, the relative-rank target shows the same qualitative separation under a different prediction-to-decision label construction. Reactive short remains the offline-selected model, while Calibrated baseline becomes the deployed-utility winner once switching costs are introduced, so we report this variant as an appendix prediction-to-decision check. The alternate $m = 0.15$ variant is also strong in the full report but is omitted from the compact table to keep the appendix narrow.

C. EPA Residual-Warning Audit

This appendix reports the full residual-candidate audit from which the expanded EPA AQS PM2.5 row is promoted to the main report card. We include all five residual candidates rather than only the Green rows. The original 16-unit EPA row is retained as prior audit context; the main text uses the expanded EPA confirmation row. Units in the expanded EPA audit are treated as report-card evaluation cells rather than independent iid samples, with leave-one-site-out and site, unit, and time-block concentration diagnostics reported in the audit artifacts.

Candidate	Verdict	κ^*	Units	$\kappa = 0$ subopt.	Subopt.	Mean shortfall	Median shortfall	CI	Placement
EPA AQS PM2.5 expanded	Green	1.00	825	0.000	0.873	17.939	12.000	[16.560, 19.342]	Main row; concentration pass
Capital Bikeshare	Green	1.00	16	0.063	1.000	2791.750	2814.000	[2743.998, 2831.502]	Appendix support
Wikimedia hourly	Green	1.00	11	0.000	1.000	118.182	115.000	[94.818, 139.727]	Appendix robustness
NYC TLC taxi-zone	Red	–	16	0.750	1.000	2285.812	2298.000	[2255.186, 2315.563]	Failed gate
M5 store-quota	Red	–	20	0.500	0.900	24.150	21.000	[17.400, 31.250]	Failed gate

Table A11. Residual-candidate audit. EPA AQS uses the expanded confirmation row; the other rows use the original residual-candidate audit. Capital and Wikimedia remain appendix-only Green support, while NYC TLC and M5 are retained as failed-gate context.

The failed candidates are not negative results for the report-card protocol. They identify cases where positive-friction gaps exist but the zero-friction alignment gate fails, which makes the result better interpreted as objective/interface mismatch rather than friction-induced offline-to-deployed selection mismatch.

D. Inventory Replenishment Corroboration and Fairness Context

D.1. Five-Family Inventory Corroboration

This appendix records the full five-family inventory summary that underlies the bounded main-text inventory evidence. The mixed zero and low-friction rows are reported here, while the main text uses only the moderate-to-high friction rows to avoid overstating the zero-friction story.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Small MLP	0.77	0.025	23/100
0.25	Small MLP	Small MLP	0.76	0.039	24/100
0.50	Small MLP	Moving average (7)	0.26	0.240	74/100
1.00	Small MLP	Moving average (7)	0.01	1.029	99/100

Table A12. Inventory corroboration under a fixed replenishment interface. The zero/low-friction regime is mixed, while moderate-to-high friction shows recurrent deployed-utility misselection.

D.2. Expanded-Family Inventory Fairness Context

The stronger-baseline sweep is included as a fairness check for the inventory corroboration. It contextualizes the inventory corroboration; the main claim is supported by the fixed-interface selection comparison in the main text.

Domain	Role	Status	Seeds	Zero flip	Zero rho	Best +fric.	Best +flip	Strongest pair	Flip share	Note
Synthetic	required	clears stricter gate	20	0.000	1.000	0.2500	0.500	Linear AR / Reactive heuristic	1.000	passed stricter gate
Inventory	required	mixed zero row	100	0.105	0.870	1.0000	0.467	Moving average (7) / Naive persistence	1.000	does not clear stricter gate

Table A13. Expanded same-interface inventory sweep. The table records broader-family status checks for inventory robustness.

Detailed tuning grids and representative selection are summarized compactly; the robustness table reports the resulting fixed-interface comparison.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Deployed-suboptimal seeds / total
0.00	Small MLP	Gradient-boosted trees	0.60	0.043	4/10
0.25	Small MLP	Gradient-boosted trees	0.60	0.059	4/10
0.50	Small MLP	Gradient-boosted trees	0.50	0.128	5/10
1.00	Small MLP	Moving average (7)	0.10	0.684	9/10

Table A14. Expanded-family inventory selection-transfer robustness summary for the broader comparison set.

The mixed zero/low-friction rows do not overturn the narrower main claim from the five-family inventory comparison.

E. Load-Following Appendix-Only Secondary Check

Load-following is a secondary operational check. It tests whether offline and deployed-utility selection can diverge outside the event-micro and Traffic-Hourly settings. Its evaluation units are disjoint client groups rather than independent random

seeds.

Friction	Forecast-side winner	Deployed winner	Agreement rate	Mean deployed gap	Median deployed gap	Deployed-suboptimal groups / total
0.00	Linear AR	Small MLP	0.70	0.000	0.000	3/10
0.25	Linear AR	Small MLP	0.30	0.001	0.000	7/10
0.50	Linear AR	Small MLP	0.30	0.001	0.001	7/10
1.00	Linear AR	Small MLP	0.20	0.002	0.002	8/10

Table A15. Load-following secondary check over disjoint client-group evaluation units. Near-zero friction remains mixed, while positive-friction regimes can favor a different deployed-utility winner.

The result is consistent with the narrower reading used throughout the paper: deployed-utility selection can differ from offline validation selection under friction, but this domain is not used as primary evidence.

F. Mechanism Check under Fixed Action Paths

Q1 serves only as mechanism support for why selection-transfer divergence can arise. Holding the proposed action path fixed, synthetic and inventory show how a fixed interface can separate proposed and realized actions once friction is introduced.

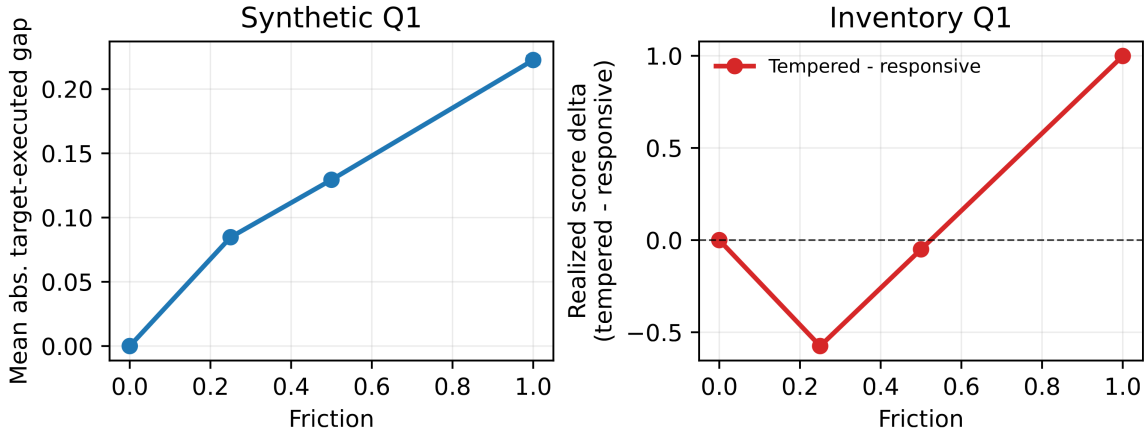


Figure A3. Q1 mechanism support. Even with the same proposed actions, a fixed frictional interface can alter realized actions and realized score; this explains how fixed-interface selection-transfer divergence can arise.

G. Uncertainty and ϵ -Tie Audit

This appendix reports uncertainty for the seed-level recurrence and paired deployed-shortfall summaries used in the main report card. We emphasize intervals and paired shortfalls rather than hypothesis-test significance. For continuous deployed shortfalls, we use 10,000 paired seed bootstrap resamples. For binary share summaries, rows with fewer than 30 matched seeds use Clopper–Pearson exact intervals; rows with at least 30 matched seeds use Wilson intervals. In the Load-following rows, the seed field denotes disjoint client-group evaluation units. We report these intervals as descriptive uncertainty summaries for the reporting diagnostic, not as hypothesis-test evidence for a broader population-level claim.

Domain	Interface	κ	Seeds	Agree.	Subopt. share/count	Subopt. 95% CI	CI type
Event-micro	threshold $\tau = 0.55$	0.50	100	0.31	0.69 (69/100)	[0.594, 0.772]	Wilson
Event-micro	threshold $\tau = 0.55$	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Traffic Top- k	budget $k = 249$	0.50	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Traffic Top- k	budget $k = 249$	1.00	100	0.00	1.00 (100/100)	[0.963, 1.000]	Wilson
Inventory	replenishment	0.50	100	0.26	0.74 (74/100)	[0.646, 0.816]	Wilson
Inventory	replenishment	1.00	100	0.01	0.99 (99/100)	[0.946, 0.998]	Wilson
Load-following	secondary	0.25	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper-Pearson
Load-following	secondary	0.50	10	0.30	0.70 (7/10)	[0.348, 0.933]	Clopper-Pearson
Load-following	secondary	1.00	10	0.20	0.80 (8/10)	[0.444, 0.975]	Clopper-Pearson

Table A16. Seed-level recurrence and share uncertainty for deployed-suboptimality. Agreement rate is the fraction of matched seeds for which the offline-selected model is also the deployed-utility best model; deployed-suboptimal share is the complementary fraction. We report Clopper-Pearson exact intervals for rows with $n < 30$ and Wilson intervals otherwise.

Aggregate recurrence intervals and uncertainty bands summarize the seed-level pattern without the dense seed-strip visualization.

Domain	Friction	Mean deployed gap	Mean gap 95% bootstrap CI	Median deployed gap	Median gap 95% bootstrap CI
Event micro	0.50	0.011	[0.009, 0.014]	0.008	[0.004, 0.011]
Event micro	1.00	0.057	[0.052, 0.063]	0.053	[0.048, 0.059]
Traffic-Hourly Top-k	0.50	7.016	[6.756, 7.283]	6.977	[6.706, 7.281]
Traffic-Hourly Top-k	1.00	24.651	[24.226, 25.079]	24.473	[24.001, 25.051]
Inventory	0.50	0.240	[0.195, 0.286]	0.202	[0.154, 0.256]
Inventory	1.00	1.029	[0.946, 1.111]	1.045	[0.926, 1.167]
Load-following	0.25	0.001	[0.000, 0.001]	0.000	[0.000, 0.001]
Load-following	0.50	0.001	[0.000, 0.002]	0.001	[0.000, 0.002]
Load-following	1.00	0.002	[0.001, 0.005]	0.002	[0.000, 0.003]

Table A17. Paired-bootstrap intervals for deployed-shortfall rows. Deployed shortfall is the paired deployed-utility shortfall $U_{\text{best}} - U_{\text{offline-selected}}$ computed within each matched seed. Intervals use 10,000 paired seed bootstrap resamples. Event-micro and Traffic-Hourly provide the main prediction-to-decision selection-transfer evidence; inventory is operational corroboration; load-following remains appendix-only directional context.

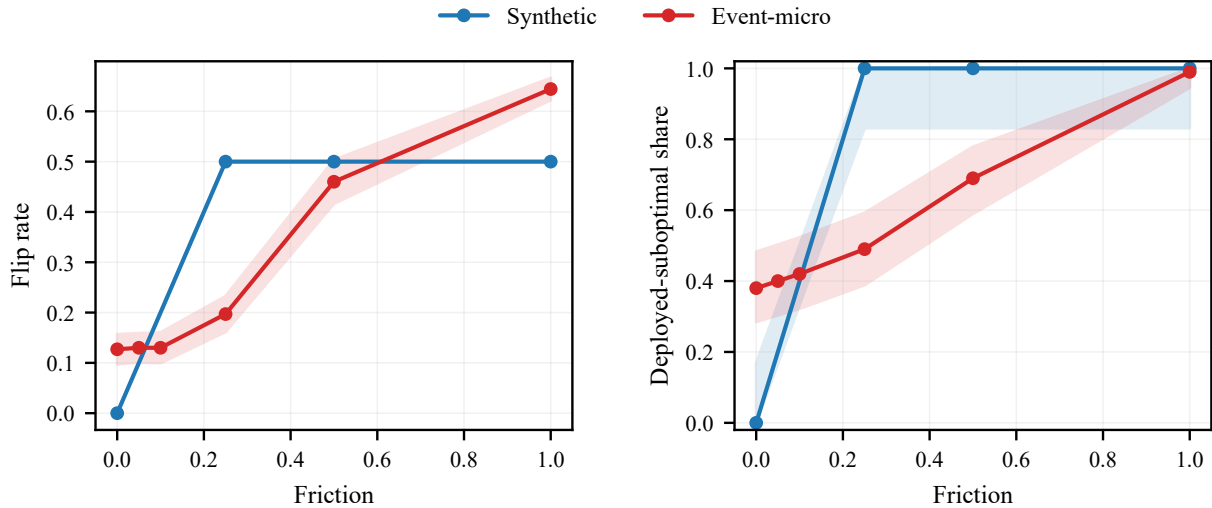


Figure A4. Selection-transfer uncertainty summaries for synthetic and event-micro. Left: flip rate; right: deployed-suboptimal share. Bands follow the interval convention described in Appendix G; zero-width bands may coincide with the line.

G.1. Epsilon-Tie Sensitivity

The audit covers Event-micro, Traffic-Hourly Top- k , Inventory, and the appendix-only Load-following check; Load-following remains appendix-only because its lower-margin rows are tie-sensitive or partially sensitive.

Table A18. ϵ -tie sensitivity for the gated report-card domains. A seed is tie-compatible when the deployed shortfall of the offline-selected model is at most $\epsilon_{\text{rel}}S + 10^{-12}$, where S is the median seed-level deployed-utility scale within the row.

Domain	Interface	κ	ϵ_{rel}	n	S	Nom. subopt.	Adj. subopt.	Tie-comp.	Strict	Median Δ	Mean Δ	Verdict
Event-micro	threshold $\tau = 0.55$	0.50	0.000	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	0.50	0.001	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	0.50	0.005	100	0.127	0.69	0.69	31	69	0.008	0.011	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.000	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.001	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Event-micro	threshold $\tau = 0.55$	1.00	0.005	100	0.085	0.99	0.99	1	99	0.053	0.057	stable
Traffic Top- k	$k=249$	0.50	0.000	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top- k	$k=249$	0.50	0.001	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top- k	$k=249$	0.50	0.005	100	12.302	1.00	1.00	0	100	6.977	7.016	stable
Traffic Top- k	$k=249$	1.00	0.000	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top- k	$k=249$	1.00	0.001	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Traffic Top- k	$k=249$	1.00	0.005	100	72.269	1.00	1.00	0	100	24.473	24.651	stable
Inventory	replenishment	0.50	0.000	100	4.841	0.74	0.74	26	74	0.202	0.240	stable
Inventory	replenishment	0.50	0.001	100	4.841	0.74	0.73	27	73	0.202	0.240	stable
Inventory	replenishment	0.50	0.005	100	4.841	0.74	0.72	28	72	0.202	0.240	stable
Inventory	replenishment	1.00	0.000	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.001	100	7.141	0.99	0.99	1	99	1.045	1.029	stable
Inventory	replenishment	1.00	0.005	100	7.141	0.99	0.99	1	99	1.045	1.029	stable

Table A19. ϵ -tie sensitivity for the appendix-only Load-following secondary check. These rows are reported separately because Load-following is not used as primary evidence and includes lower-margin tie-sensitive rows.

Domain	Interface	κ	ϵ_{rel}	n	S	Nom. subopt.	Adj. subopt.	Tie-comp.	Strict	Median Δ	Mean Δ	Verdict
Load-following	responsive	0.25	0.000	10	0.046	0.70	0.70	3	7	0.000	0.001	tie-sensitive
Load-following	responsive	0.25	0.001	10	0.046	0.70	0.70	3	7	0.000	0.001	tie-sensitive
Load-following	responsive	0.25	0.005	10	0.046	0.70	0.50	5	5	0.000	0.001	tie-sensitive
Load-following	responsive	0.50	0.000	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	0.50	0.001	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	0.50	0.005	10	0.058	0.70	0.70	3	7	0.001	0.001	stable
Load-following	responsive	1.00	0.000	10	0.081	0.80	0.80	2	8	0.002	0.002	partially sensitive
Load-following	responsive	1.00	0.001	10	0.081	0.80	0.80	2	8	0.002	0.002	partially sensitive
Load-following	responsive	1.00	0.005	10	0.081	0.80	0.70	3	7	0.002	0.002	partially sensitive